

In every icon the grey track is GENCODE; the coloured track is what flightcollapse emits. Faint dashed boxes are reference exons the model does not contain.
0.1.18 changes the 3' end only — panel 5. Annotation may now refine a read-supported cleavage coordinate but never manufacture a distant one, and the polyA tail is read from each molecule instead of inferred from the genome.

1 Read admission

One definition of an admitted read, shared by the scanning pass and the output pass, so the two can never disagree.

admitted

rejected

coverage ≥ 0.90 · identity ≥ 0.95 · 5' soft clip ≤ 200 nt

On an untrimmed BAM the polyA tail is soft-clipped, so a 0.99 coverage gate keeps only 6% of reads — and enriches for mono-exonic ones.

2 Junction census

Every junction in every admitted read is counted once, with its motif, its flanking repeat, its share of the traffic at each of its own splice sites, and its distance to the nearest annotated site.

GENCODE

read

snapped

annotation-first: a read site within 5 bp moves onto the annotated site

internal deletions are promoted first

A short intron spelled as a CIGAR D rather than an N leaves the intron chain untouched, so a real splicing event can hide inside an "FSM". Gaps of 20–5,000 bp seen in ≥ 3 reads are promoted to junctions before the census, then curated like any other.

does the read have any intron?

YES → spliced track

NO → mono-exon track, panel 6

The empty chain is not a suffix of anything. That single rule is the mono-exon fix.

→ unspliced reads skip panels 3-5 entirely

3 Junction curation

Junctions are curated BEFORE chains are built. A chain with 40 reads containing one spurious junction survives a chain threshold; it does not survive a junction threshold.

An ANNOTATED junction is never rejected — dropping one because it is rare in one sample would fabricate novel chains out of well-known transcripts.

a junction that is still novel must clear

- molecules ≥ 3 and reads ≥ 5

not read count alone — reads are PCR copies

- ≥ 0.1% of the reads at its own donor AND its own acceptor

was 1%. That threshold rejected 174 of 230 real junctions the caller lost, and the ones it rejected carried 51 reads and 49.5 cells at the median against 13 and 12 for the novel junctions it kept.

- GT-AG always; GC-AG and AT-AC only with short-read confirmation in ≥ 2 replicates, or ≥ 50 reads

was GT-AG only. That cost 8–10% of real junctions — including SOD1's canonical transcript: GC-AG, 10,419 long reads, 1,005 short reads in 3 of 3 replicates.

- no direct repeat ≥ 8 bp flanking the intron

RT template switching

- local FDR ≤ 0.05 from the calibrated model

see below

the calibrated model the last bullet refers to

features

support · cells · reads-per-molecule · donor share · acceptor share · intron length · direct repeat · motif class · distance to annotated sites

Ridge logistic → two-component mixture → local FDR, with Storey π_0 and a monotone fit, stratified by whether the junction reuses an annotated splice site. Annotated junctions are the positive anchor; junctions on rejected reads are the negative one.

lfdr, posterior_real_in_junctions.tsv

why those two gates moved

Share and motif are already features of that model. It scored the junctions the two hard gates were rejecting at IFDR 0.000 — confidently real — and the thresholds overruled it on its own inputs. That is what 0.1.16 removes: not a loosening, a de-duplication.

a read carrying a rejected junction is set aside

read

rejected junction

It never becomes a model — and it becomes the negative anchor the chain-level model trains on. Relaxing the gates shrinks that set as a side effect, and an empty decoy set makes the chain calibration fall back to flat thresholds silently, so 0.1.16 counts decoy groups and fails the run below a floor.

decoy_chain_groups in qc.json

short reads are an anchor, never a filter

Short reads may SAVE a junction and can never condemn one. At 24% unique mapping, "the short reads did not see it" is not evidence of absence — 227 of 525 known-real junctions were invisible in all three replicates.

rescue is opt-in: junctions.short_read_rescue

4 Chain identity

Reads are grouped on the exact curated intron chain, stored in transcript orientation so that a 5'-truncated read is always a SUFFIX of its parent.

chain matches an annotated chain

GENCODE

model

FSM — the model takes that transcript's GENCODE identity.

ENST00000229239.10

unannotated chain that is a SUFFIX of an annotated one

GENCODE

reads

model

5' truncation — merged into the parent whatever the support ratio says. The truncated form routinely outnumbered the intact one here.

...unless its 5' end sits at an annotated TSS

GENCODE

model

A real alternative start — kept as its own model.

ENST...|alt5end1

independent chain, every junction annotated

GENCODE

model

NIC — a novel combination of known splice sites.

GAPDH|NIC_1

independent chain carrying a novel junction

GENCODE

model

NNC — reaches this point only if that junction survived stage 3.

GAPDH|NNC_1

an intron is retained, with exonic flank on BOTH sides

GENCODE

model

IR. Flank on both sides is what separates a retained intron from a plain 3' truncation that happens to stop inside one.

ENST...|IR1

the chain spans two annotated genes

GENCODE

model

READTHROUGH — its own category, not a novel isoform of either gene.

GENEA|readthrough1

neither side annotated and support is comparable

chain A

chain B

Genuinely ambiguous — kept separate and flagged, never guessed.

flagged in group_chains.tsv

The representative is ALWAYS the maximal chain: merges only ever run from a shorter chain into a longer one.

5 Terminal ends

3' is the anchored end under oligo-dT, so it is peak-clustered. 5' is the degraded end: a model's 5' end is the 90th percentile of its members, never the minimum.

Ends are compared against EVERY transcript sharing the chain, not against one designated parent.

new in 0.1.18 · a PCR family no longer picks the site

Peaks are found, ranked and placed on MOLECULE mass — each read weighted 1/(reads sharing its cell+UMI). Support gates still count whole reads and whole molecules; only the geometry changed. On an already deduplicated BAM every weight is 1 and this is a no-op.

weight_ends_by_molecule

new in 0.1.18 · polyA evidence comes from the molecule

The 3' soft clip is read as SEQUENCE, not just measured. A template-free A-run outside the alignment (poly-T on a minus-strand read) is the only per-molecule evidence that this coordinate is a cleavage site. Median 96% of a model's molecules carry one; models flagged for internal priming carry a 12 bp tail against 28 bp elsewhere.

median_tail_len, tail_molecule_frac in models.tsv · polya_tail_gates_novel_end

peak matches the parent transcript's annotated 3' end

GENCODE

model

FSM.

ENST00000229239.10

peak matches ANOTHER transcript of the same gene

GENCODE

model

3' end of tx B

An annotated alternative end, not a discovery. Checking only the parent transcript undercounts badly: 35% of 3' ends match it, 72% match some transcript of the gene.

ENST...|alt3end1

novel peak, ≥ 5% of the group, polyA tail on its molecules

GENCODE

model

AAAA...tail

A genuine alternative polyA site. In 0.1.16 the evidence was genomic — a PAS hexamer and downstream A-content, both statements about the LOCUS. The tail is a statement about the molecule that was primed there, and it is what decides now.

ENST...|novel3end1

peak rejected, but an annotated end is within 300 bp

GENCODE

model

≤ 300 bp

Snapped. This is annotation REFINING an observed coordinate, and it stays: the median model still moves 6 bp and no model moves more than 300.

FSM

peak rejected and the nearest annotated end is far away

read 3' ends

0.1.16

0.1.18

moved 11.9 kb

0.1.16 re-labelled the peak with the parent's annotated end at ANY distance — at NSL1 that put the model 11.9 kb past every read supporting it, and it happened to 19,145 peaks per sample at a median of 1.2 kb. The observed coordinate now stands and the model says it is unresolved.

ENST...|end3_unresolved_1

A-rich genomic window downstream, no tail on the reads

model

A-rich, no tail

Still discarded as internal priming. The oligo-dT primes on genomic A's, which ALIGN rather than clip — so an internally primed molecule shows a read pile and no tail. That is the one thing the read pile alone cannot tell you.

5' end upstream of the annotated TSS

GENCODE

model

A credible alternative start — unless it crosses an annotated junction, which makes it a retained first intron instead. A 5' truncation, by contrast, is degradation and is absorbed into the parent.

ENST...|ext5end1

parent_transcript — identical chain, nothing weaker

A PARENT is only ever a GENCODE transcript whose intron chain equals the model's, within the ±5 bp fuzzy tolerance. Every terminal-end category has one by construction. NIC and NNC have a novel chain, so they have NO parent — parent_transcript is empty for 100% of NIC and NNC models.

From chains.py: ref.transcripts_for_chain(). Among those, the one whose annotated 3' end is nearest the dominant peak.

associated_transcript — merely the closest

Separately, EVERY model is matched to its nearest reference transcript — scored on shared, model-only and reference-only junctions, then [3' distance], then [5' distance] — as associated_transcript. Where both exist they agree 98.7% of the time, which is why conflating them looks harmless until a NIC or NNC shows up.

Annotation only. It never feeds back into calibration, and it never changes which model is emitted. A model NAME falls back to it when there is no parent, so ENST...|NNC_1 names a neighbour, not a parent.

6 Mono-exon track

Runs entirely separately. Unspliced reads only ever group with each other, so a single-exon stub can never absorb spliced read mass.

inside the last exon or 3'UTR of a known gene

GENCODE

model

Trusted: with polyA capture this is RT drop-off in a long 3'UTR, and the 3' end is still correct.

ENST...|3UTRfrag1

inside a gene body, not terminal

GENCODE

model

AATAAA

Needs an actual polyA site — otherwise discarded.

GENE|APA1

intergenic

GENCODE

model

AATAAA

Needs a polyA site and a higher support floor.

NOVELMONO_chr1_3

A-rich genomic window downstream of the 3' end

model

%A ≥ 60

Internal priming — discarded in every case. Note the 0–100 scale: comparing this to 0.6 flags essentially everything.

7 Invariants — checked every run

- mono-exon read mass ≈ unspliced read fraction

9.2% of reads unspliced against 57.9% of read mass on single-exon models is how the isoseq bug was found.

- model_is_min must not be 1.0

If the model is always the shortest member of its own group, the representative rule is inverted.

- median log2(read span / model span) = 0

Reads should match the footprint of the model they were assigned to.

- unspliced reads are accounted for

Reads that vanish are as much a modelling failure as reads that pile onto a stub — just a quieter one.

- the decoy set has not collapsed

Below 2,000 decoy chain groups the chain-level model has no negative anchor and reverts to flat thresholds — silently, until 0.1.16 made it fail.

- rejected junctions carry LESS evidence than kept ones

qc.md prints rejections per rule with the median reads and IFDR of each. A rule discarding junctions at IFDR 0.000 is visible on any run instead of taking a masking experiment to find.

- no 3' end has moved further than the bound

New in 0.1.18. qc.md reports max_abs_tts_shift; it is 300 by construction, so a larger value means a second code path is moving ends and is a bug.

- the polyA tail was actually measured

New in 0.1.18. A hard-clipped or trimmed BAM yields zero tails, which is indistinguishable from 'no molecule had one' — so the report says which.